

Part 1: Exploring the SANS Website

Step 1 – Three resources on the SANS FREE Resources page

One resource available is @RISK: The Consensus Security Alert, which is a weekly newsletter that summarizes the most serious new vulnerabilities and malware threats. It helps security professionals quickly understand what issues need immediate attention.

Another resource is the CIS Critical Security Controls, which provide a prioritized list of cybersecurity best practices that organizations can follow to improve their overall security posture.

The third resource is the SANS Reading Room, which contains a large collection of white papers and research articles on various information security topics written by security professionals.

Step 2 – One CIS Critical Security Control and how it can be implemented

One important control is Inventory and Control of Hardware Assets.

To implement this control, organizations should actively scan their networks to identify all connected devices. They should maintain an updated asset inventory database and use tools like DHCP logs to automatically track new devices. Network Access Control (NAC) systems can be used to prevent unauthorized devices from connecting. Regular audits should also be performed to compare detected devices with the official inventory list.

Step 3 – Three SANS Newsletters

1. @RISK: The Consensus Security Alert – A weekly newsletter that ranks the most critical newly discovered vulnerabilities and malware threats.
2. SANS NewsBites – A twice-weekly summary of major cybersecurity news stories, often including commentary from SANS experts.
3. OUCH! – A monthly newsletter written for everyday users that focuses on security awareness topics like phishing, password safety, and online privacy.

Part 2: Identify Recent Network Security Threats

Step 1 – Examples of recent vulnerabilities mentioned in @RISK

Recent threats highlighted in security alerts include: Remote code execution vulnerabilities affecting unpatched Microsoft Windows systems through SMB, Critical browser vulnerabilities in Chrome and Firefox that allow sandbox escapes, SQL injection flaws in popular web applications and WordPress plugins, Zero-day vulnerabilities targeting Adobe Acrobat and Reader, Authentication bypass vulnerabilities in VPN devices such as Pulse Secure and Fortinet, Privilege escalation vulnerabilities in Linux kernels

Step 2 – Other websites that provide security threat information

Several reliable websites provide updated information about cybersecurity threats: US-CERT, National Vulnerability Database, Krebs on Security, Threatpost, Threat Intelligence, Kaspersky SecureList

Recent threats reported on these sites include ransomware attacks targeting healthcare organizations, supply chain attacks similar to SolarWinds, phishing campaigns impersonating Microsoft 365 and Google login pages, DDoS attacks using botnets like Mirai variants, and credential stuffing attacks targeting financial and e-commerce platforms.

Part 3: Detailed Network Security Attack

Attack: WannaCry Ransomware

Name of Attack: WannaCry Ransomware

Type of Attack: Ransomware / Worm

Dates of Attack: May 12–15, 2017 (major outbreak)

Who Was Affected:

WannaCry infected over 200,000 computers in more than 150 countries. Major organizations impacted included the UK's National Health Service (NHS), FedEx, Telefónica, and Deutsche Bahn.

How the Attack Worked:

WannaCry exploited a vulnerability in the Windows SMB protocol known as EternalBlue (CVE-2017-0144). This exploit allowed the malware to spread automatically across networks without requiring users to click anything. Once infected, files on the system were encrypted and victims were asked to pay a Bitcoin ransom (usually between \$300 and \$600) to regain access. Systems that had not installed Microsoft's MS17-010 security patch were especially vulnerable.

Mitigation Strategies:

Organizations can reduce risk by installing security updates immediately, disabling the outdated SMBv1 protocol, blocking ports 139 and 445 when possible, using network segmentation, maintaining secure offline backups, and deploying endpoint detection and response tools. Keeping antivirus software updated is also important.

Reflection Questions

1. Steps to protect your own computer

To protect my own computer, I would make sure my operating system and applications are always updated. I would use strong, unique passwords and store them in a password manager. Enabling multi-factor authentication on important accounts adds another layer of protection. I would also back up my data regularly, avoid suspicious email attachments and links, use antivirus software, and avoid unsecured public Wi-Fi networks (or use a VPN when necessary).

2. Important steps organizations can take

Organizations should use a layered security approach. This includes regular vulnerability scans, strong patch management policies, employee security awareness training, network segmentation, and enforcing the principle of least privilege. They should also maintain an incident response plan, continuously monitor their networks through a SIEM or SOC, and store encrypted backups offline. Following recognized frameworks such as CIS Controls or the NIST Cybersecurity Framework can help ensure best practices are consistently applied.